Siberiana: how to present online lightly digitized archaeological cultures of Yenisei Siberia

Volodin, Andrey
volodin@hist.msu.ru
Siberian Federal University, Russian Federation; Moscow Lomonosov University, Russian Federation

Senotrusova, Polina
psenotrusova@sfu-kras.ru
Siberian Federal University, Russian Federation

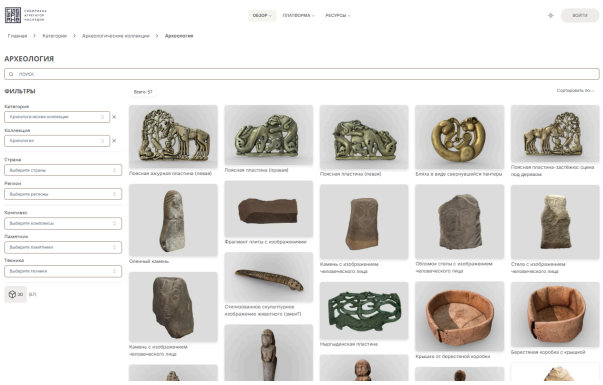
Rumyantzev, Maksim
mrumyantsev@sfu-kras.ru
Siberian Federal University, Russian Federation

Pikov, Nikita
npikov@sfu-kras.ru
Siberian Federal University, Russian Federation

Kizhner, Inna
inna.kizhner@gmail.com
Haifa University, Israel

In 2024, Digital Humanities Research Institute at Siberian Federal University develops the digital research infrastructure for the aggregation, preservation, dissemination of Siberian historical and cultural heritage — Siberiana.online (*or Сибириана*).

Picture 1. *Siberiana.online: aggregator home page*
< *https://siberiana.online/* >



The aim of the project is to establish a long-winded collection platform for digitization, analysis, and curation of the cultural heritage of the Central Siberia or so-called Yenisei Siberia.

To implement the project, Siberian Federal University organized a series of expeditions to museums in the huge Krasnoyarsk region (2300000 km ^{2}) for digitizing, describing and subsequently publishing various archaeological collections on the Siberiana platform.

The poster is devoted to the practical experience of creating a digital collection of archaeological heritage of Yenisei Siberia. The issues of defining description fields, data completeness, searching for necessary information are considered. The problems of completing the collections of regional and city museums are revealed.

The project is designed for research and education needs of the digital humanists at Siberian Federal University and world over, because judging by the current research Siberian artifacts and collections evoke a steady interest, although they are often underestimated due to the small number of already digitized objects (Kizhner et al, 2021).

The Siberiana is organized on the principle of a global aggregator of data text and media (including 3D-models and digital twins). The collected information already gives an opportunity to get acquainted with historical, archaeological, ethnographic and cultural artifacts from the region.

The platform has a two-level access structure. The first level is intended for a wide audience of users: content is available for free download, and can be used for research or educational purposes, as well as for self-education or sightseeing. The second level of access is addressed to professional researchers, such as historians, archaeologists, anthropologists or museum staff, and requires authorization.

Siberiana is a very large project, such a “site of sites”, reminiscent of aggregators like Europeana. The very aim is to present online most of archeo-

logical cultures from the vast territories around the Yenisei River in the heart of Siberia. The beta version of Siberiana works for free.

Today the project team is working on the creation of digitized content, overcoming such problems as the disparate museum holdings and archaeological collections, the slow spread of “digitalization” in the historical and local history sphere. Siberian Federal University has already implemented successful projects related to the digitization of Siberian and metropolitan art collections. This aim gives an extra opportunity for interaction with holders of historical and cultural heritage, for creation of thematic research collections, and it gives challenges for formulating automatic selection goals (Kizhner et al, 2021).

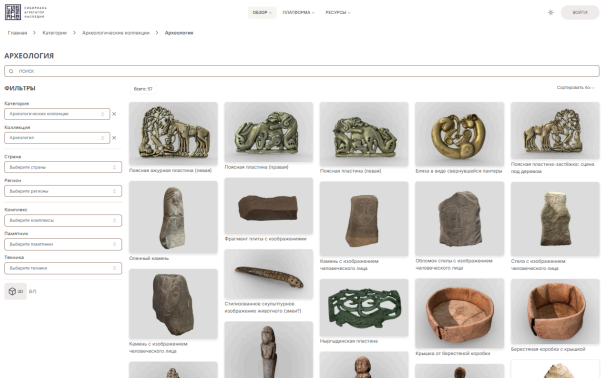
Siberiana challenges the problem of using digital twins in archaeological practice. The experience of using digital technologies in archaeological research is characterized: creation of digital models of objects, development of geoinformation systems, formation of new approaches to data storage and utilization, use of digital twins for complex research or restoration procedures. Technical and substantive problems of using digital twins in archaeology are emphasized.

The consortium of Russian Digital Humanities initiatives was created to unite museums, libraries, archives with researchers and data scientists and engineers in developing such complex tasks of *Siberiana* as the digital infrastructure of cultural heritage of the region.

The different tools have been developed for aggregation and representation of already digitized data (annotated images, electronic copies of historical documents, 3D/VR/AR/MR models). The CIDOC standard for describing cultural heritage is used thanks to the recent translation of the version 7.1 into Russian. Examples of best practices are collected that can be included and scaled for the tasks of *Siberiana*.

The Krasnoyarsk region is very rich in archaeological finds, but many of them have long remained unexplored in the vaults of museums. That’s why for the initial stage of *Siberiana*, archeological collections were selected. These collections are quite complete, amenable to a complete description. At the same time, they are diverse, and there’s a demand from researches to study them.

Picture 2. *Archaeological artifacts presented at Siberiana.online*
(2446 items *)*
< *https://siberiana.online/objects?category=archaeological-collections* >



Picture 3. *3-dimentional models of archaeological artifacts presented at Siberiana.online (* 57 items *)*
< *https://siberiana.online/objects?category=archaeological-collections&type=artifacts&modelArtifacts=true* >

Siberiana, is structured as a dataset with faceted classification of the obligatory properties are object type, time period, location, institution owner, as well as copyright or open license information.

Siberiana is plotting as a digital model of the historical, cultural, and natural heritage of the Yenisei region. Our goal is to unlock the potential of the heritage resources in two dimensions. For all types of users: it's to make such content easily accessible, visually well-presented, conveniently organized, and for researchers: it’s to provide a wide coverage of collections, valuable services for exploratory data analysis, to give a platform for collecting specific research data for a significant agenda (Kizhner et al, 2022).

Thus, historical, cultural, and natural heritage should be understood as a resource that allows people to make life interesting, rich, valuable, and in some respects socially significant.

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